

A Practical Guide to Stable Diffusion

TOPIC -- STABLE DIFFUSION

$$\mathcal{L} = \mathbb{E}_{x, \epsilon, t} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]$$

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Stable Diffusion Demo "Step by Step visual introduction to Diffusion Models. - Blog by Kemal Erdem". Retrieved August 31, 2024. "U-Net for Stable Diffusion". U-Net for Stable Diffusion. Retrieved August 31, 2024. Interactive Explanation of Stable Diffusion "We Are All Raw Material for AI": Investigation on sensitive and private data in Stable Diffusions training data

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SD XL The XL version uses the same LDM architecture as previous versions, except larger: larger UNet backbone, larger cross-attention context, two text encoders instead of one, and trained on multiple aspect ratios (not just the square aspect ratio like previous versions). The SD XL Refiner, released at the same time, has the same architecture as SD XL, but it was trained for adding fine details to preexisting images via text-conditional img2img.

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Andersen, McKernan, and Ortiz v. Stability AI, Midjourney, and DeviantArt In January 2023, three artists, Sarah Andersen, Kelly McKernan, and Karla Ortiz, filed a copyright infringement lawsuit against Stability AI, Midjourney, and DeviantArt, claiming that these companies have infringed the rights of millions of artists by training AI tools on five billion images scraped from the web without the consent of the original artists. In July 2023, U.S. District Judge William Orrick inclined to dismiss most of the lawsuit filed by Andersen, McKernan, and Ortiz but allowed them to file a new complaint, providing them an opportunity to reframe their arguments.

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The 3.0 version completely changes the backbone. Not a UNet, but a Rectified Flow Transformer, which implements the rectified flow method with a Transformer. The Transformer architecture used for SD 3.0 has three "tracks", for original text encoding, transformed text encoding, and image encoding (in latent space). The transformed text encoding and image encoding are mixed

during each transformer block. The architecture is named "multimodal diffusion transformer (MMDiT)", where the "multimodal" means that it mixes text and image encodings inside its operations. This differs from previous versions of DiT, where the text encoding affects the image encoding, but not vice versa.

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ControlNet ControlNet is a neural network architecture designed to manage diffusion models by incorporating additional conditions. It duplicates the weights of neural network blocks into a "locked" copy and a "trainable" copy. The "trainable" copy learns the desired condition, while the "locked" copy preserves the original model. This approach ensures that training with small datasets of image pairs does not compromise the integrity of production-ready diffusion models. The "zero convolution" is a 1×1 convolution with both weight and bias initialized to zero. Before training, all zero convolutions produce zero output, preventing any distortion caused by ControlNet. No layer is trained from scratch; the process is still fine-tuning, keeping the original model secure. This method enables training on small-scale or even personal devices. ControlNet is commonly used to alter the generation of an image based on an input image that maps properties desired in the final image. Common mapping image types include a depth map, edges, or one or more skeletal poses. These inputs may be generated directly, but they are often derived from still other images, using neural nets or processes like edge detection.

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Getty Images v. Stability AI In January 2023, Getty Images initiated legal proceedings against Stability AI in the English High Court, alleging significant infringement of its intellectual property rights. Getty Images claims that Stability AI "scraped" millions of images from Getty's websites without consent and used these images to train and develop its deep-learning Stable Diffusion model. Key points of the lawsuit included:

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Usage and controversy Stable Diffusion claims no rights on generated images and freely gives users the rights of usage to any generated images from the model provided that the image content is not illegal or harmful to individuals. The images Stable Diffusion was trained on have been filtered without human input, leading to some harmful images and large amounts of private and sensitive information appearing in the training data. More traditional visual artists have expressed concern that widespread usage of image synthesis software such as Stable Diffusion may eventually lead to human artists, along with photographers,

models, cinematographers, and actors, gradually losing commercial viability against AI-based competitors. Stable Diffusion is notably more permissive in the types of content users may generate, such as violent or sexually explicit imagery, in comparison to other commercial products based on generative AI. Addressing the concerns that the model may be used for abusive purposes, CEO of Stability AI, Emad Mostaque, argues that "[it is] peoples' responsibility as to whether they are ethical, moral, and legal in how they operate this technology", and that putting the capabilities of Stable Diffusion into the hands of the public would result in the technology providing a net benefit, in spite of the potential negative consequences. In addition, Mostaque argues that the intention behind the open availability of Stable Diffusion is to end corporate control and dominance over such technologies, who have previously only developed closed AI systems for image synthesis. This is reflected by the fact that any restrictions Stability AI places on the content that users may generate can easily be bypassed due to the availability of the source code. Controversy around photorealistic sexualized depictions of underage characters have been brought up, due to such images generated by Stable Diffusion being shared on websites such as Pixiv. In June of 2024, a hack on an extension of ComfyUI, a user interface for Stable Diffusion, took place, with the hackers claiming they targeted users who committed "one of our sins", which included AI-art generation, art theft, promoting cryptocurrency.

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Stable Diffusion also includes another sampling script, "img2img", which consumes a text prompt, path to an existing image, and strength value between 0.0 and 1.0. The script outputs a new image based on the original image that also features elements provided within the text prompt. The strength value denotes the amount of noise added to the output image. A higher strength value produces more variation within the image but may produce an image that is not semantically consistent with the prompt provided. There are different methods for performing img2img. The main method is SDEdit, which first adds noise to an image, then denoises it as usual in text2img. The ability of img2img to add noise to the original image makes it potentially useful for data anonymization and data augmentation, in which the visual features of image data are changed and anonymized. The same process may also be useful for image upscaling, in which the resolution of an image is increased, with more detail potentially being added to the image. Additionally, Stable Diffusion has been experimented with as a tool for image compression. Compared to JPEG and WebP, the recent methods used for image compression in Stable Diffusion face limitations in preserving small text and faces. Additional use-cases for image modification via img2img are offered by numerous front-end implementations of the Stable

Diffusion model. Inpainting involves selectively modifying a portion of an existing image delineated by a user-provided layer mask, which fills the masked space with newly generated content based on the provided prompt. A dedicated model specifically fine-tuned for inpainting use-cases was created by Stability AI alongside the release of Stable Diffusion 2.0. Conversely, outpainting extends an image beyond its original dimensions, filling the previously empty space with content generated based on the provided prompt. A depth-guided model, named "depth2img", was introduced with the release of Stable Diffusion 2.0 on November 24, 2022; this model infers the depth of the provided input image, and generates a new output image based on both the text prompt and the depth information, which allows the coherence and depth of the original input image to be maintained in the generated output.